

Social Engine Data Intake Restoration & EDA Report

Event: Data Vortex — Aaruush '26 | Round 1 (Phase 1)
Author: Kislay Jha

1. Executive Summary

The Social Engine suffered a catastrophic intake breakdown resulting in record duplication, signed integer inversion, unparsed multi-dialect timestamps, and null categorical fields. Following the retrieval of Dataset 1 from surviving node_07, an end-to-end cleaning and exploratory analysis pipeline was deployed. The data was restored to 12,000 verified relational rows with zero integrity violations.

2. Ingestion Integrity Audit & Cleaning Methodology

Anomaly Class	Raw State Defect	Resolution Methodology	Justification
Duplicate Records	360 redundant rows sharing primary key <code>post_id</code> .	Deduplication preserving unique <code>post_id</code> instances.	Prevents artificial metric inflation in downstream analytical engines.
Bit-Flip Sign Inversion	525 posts with negative likes (min: -4,987; median: -2,388).	Absolute value restoration (<code>abs(x)</code>).	Negative likes mirrored positive distribution curves while shares and comments remained strictly non-negative (≥ 0).
Missing Likes	1,858 null entries.	Imputed using platform-specific median values.	Preserves metric distributions without skewing aggregate means.
Platform Incompleteness	1,846 null platform labels.	Standardized to categorical label <code>'Unknown'</code> .	Cross-platform posting patterns proved users are multi-homed (up to 5 platforms), disallowing 1:1 user inference.
Timestamp Desynchronization	Three-way collision: Unix epoch integers, ISO-8601 strings,	Multi-dialect regex parser normalizing records to unified UTC ISO datetime (<code>YYYY-MM-DD HH:MM:SS</code>).	Ensures cross-temporal aggregation accuracy in relational databases.

Anomaly Class	Raw State Defect	Resolution Methodology	Justification
	and DD-MM-YYYY dates.		
Relational Integrity	12,000 posts referencing 1,500 users.	Foreign key verification against <code>Social_Engine_Users.csv</code> .	0 orphan records; 100% relational integrity maintained.

3. Exploratory Data Analysis & System Diagnostics

A. Platform Volume & Engagement Symmetry

The platform exhibits near-uniform generation volume and symmetrical engagement:

- **Facebook:** 2,074 posts | Mean Engagement: 4,026.76
- **Instagram:** 1,989 posts | Mean Engagement: 4,040.83
- **Reddit:** 2,031 posts | Mean Engagement: 4,002.36
- **Twitter:** 2,049 posts | Mean Engagement: 3,949.26
- **YouTube:** 2,073 posts | Mean Engagement: 4,044.55
- **Unknown Platform:** 1,784 posts | Mean Engagement: 3,973.59

B. Temporal Intake Collapse (Anomaly Discovery)

Monthly analysis uncovers a critical event in the Social Engine lifecycle:

- **Normal Operations (Jan 2024 – Apr 2025):** Mean intake sustained at ~960 posts/month.
- **Intake Failure (May 2025 – Dec 2025):** Post ingestion plunged abruptly by **95.2%**, collapsing to an average of ~48 posts/month. This marks the exact timeline of node network severance.

C. Geographic Activity Distribution

Ingestion activity is globally distributed, led by:

1. **Los Angeles, USA:** 459 posts
2. **Munich, Germany:** 452 posts
3. **Shanghai, China:** 451 posts
4. **Barcelona, Spain:** 439 posts
5. **Houston, USA:** 422 posts

D. Audience Engagement Decoupling

Correlation analysis reveals a total decoupling between audience size and engagement metrics:

- $r(\text{follower_count}, \text{total_engagement}) = 0.0013$
- $r(\text{likes}, \text{shares}) = -0.0012$
- $r(\text{likes}, \text{comments}) = 0.0095$

Engagement behavior is uncorrelated with author scale, indicating an algorithmic recommendation architecture that distributes impressions independently of static follower counts.